

SPH4U Unit 3 Gravitational, Electric, and Magnetic Field

Chapter 7 Electric Fields Lesson Package

Schoolwork	Useful Links/ Videos/ Handouts	Homework
Chapter 7 Videos		
	1. (a collection of videos) Electric force – collection of videos	
7.1 Properties of Electric Charge		
1. Lesson 7.1 Presentation 2. Lesson 7.1 Lesson Package	Videos: 1. 7.1 Properties of electric charge	1. 7.1 Check Your Understanding 2. Textbook pg. 326 #1-11
7.2 Coulomb's Law		
1. Lesson 7.2 Presentation 2. Lesson 7.2 Lesson Package	Videos: 1. 7.2 Coulomb's law	1. 7.2 Check Your Understanding 2. Textbook pg. 333 #1-10
7.3 Electric Field		
1. Lesson 7.3 Presentation 2. Lesson 7.3 Lesson Package	Videos: 1. 7.3 Electric fields	1. Textbook pg. 345 #1-8
7.4 Potential Difference and Electric Potential		
1. Lesson 7.4 Presentation 2. Lesson 7.4 Lesson Package	Videos: 1. 7.4 Electric potential energy and potential difference	1. Textbook pg. 354 #1-10
7.5 Electric Potential and Electric Potential Energy Due to a Point Charge		
1. Lesson 7.5 Presentation 2. Lesson 7.5 Lesson Package	Videos: 1. 7.5 Electric potential due to point charges	1. Textbook pg. 361 #1-8

7.6 The Millikan Oil Drop Experiment		
1. Lesson 7.6 Presentation 2. Lesson 7.6 Lesson Package	Videos: 1. Millikan's oil drop experiment to determine charge of an electron 2. Millikan Oil Drop Experiment Animation 3. The Millikan Experiment 4. 7.6 The Millikan oil drop experiment	1. Textbook pg. 365 #1-8
Chapter 7 Review		
1. Chapter 7 Self- Quiz pg. 369 (attempt all) 2. Chapter 7 Review pg. 370 #1-57 3. Communication Questions (lesson package)		1. Chapter 7 Self- Quiz pg. 369 (attempt all) 2. Chapter 7 Review pg. 370 #1-57 3. Chapter 7 - Check Your Knowledge – Definitions (lesson package) 4. Communication Questions (lesson package)

Lesson 7.1 Properties of Electric Charge

Lesson Goal: How do electric charges move from one place to another and charge an object?

Success Criteria:

- ✓ explain the law of electric charges
- ✓ explain the law of conservation of charge
- ✓ describe the differences between a conductor and an insulator
- ✓ describe the different ways an object can be charged
- ✓ explain the flow of charge in each charging method

Go through the 7.1 PPP posted on Schoology and lesson 7.1 in the textbook.

Law of Electric Charge

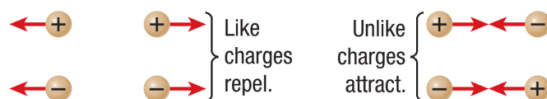


Figure 3 The electric force between two like charges (charges with the same sign) is repulsive, whereas the electric force between unlike charges (charges with opposite signs) is attractive.

Law of Conservation of Charge

Lesson Reflection

State one thing you learned in this lesson:

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State one thing you need to go over:

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Check Your Understanding - Lesson 7.1

1. What makes an object positively charged?
2. Consider four types of charging an object: friction, induced charge separation, contact, and induction. Determine, with examples, how charge is created in each process.

7.2 Coulomb's Law

Lesson Goal: How the electric force depends on charge and distance and solve problems related to the electric force.

Success Criteria:

- ✓ calculate the force between two point charges when you know the distance between them
- ✓ describe the similarities between the gravitational force and the electric force
- ✓ apply the superposition principle to calculate the total force exerted on a point charge from two or more charges

Now that we went through [PP5 \(Coulomb's Law\)](#) and [PP6 \(Using Coulomb's Law\)](#)

Electric Force

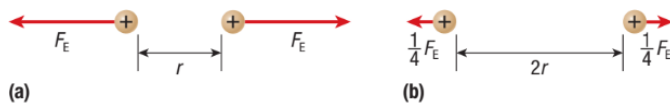


Figure 1 The electric force of repulsion between two identical charges decreases as the separation increases.

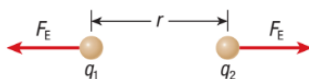
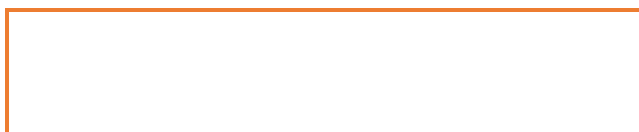


Figure 2 The electric force between two point charges q_1 and q_2 is given by Coulomb's law.

Coulomb's Law

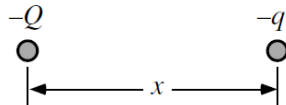
Coulomb's Law has several important properties:

- The electric force is repulsive for like charges and attractive for unlike charges. Mathematically, this property results from the product q_1q_2 in the numerator of the Coulomb's law equation. The magnitude of the electric force, F_E , is always positive. In physics, it is convenient to use a negative sign to show direction. Including the negative sign of a negatively charged object would imply a direction. **You never include the sign of the charge when solving problems related to Coulomb's law.**
- The electric force is inversely proportional to the square of the distance between the two particles.
- The magnitude of the electric force is the magnitude of the force exerted on each of the particles. That is, a force of magnitude F_E is exerted on charge q_1 by charge q_2 , and a force of equal magnitude and opposite direction is exerted on q_2 by q_1 . This pairing of equal but opposite forces is described by Newton's third law, the action–reaction principle.

Comparing Coulomb's Law and Universal Gravitation

TIPERS**D1-WWT10: TWO NEGATIVE CHARGES—FORCE**

Two negatively charged particles are separated by a distance x . The particle on the left has a charge $-Q$ which is three times the charge $-q$ of the particle on the right.



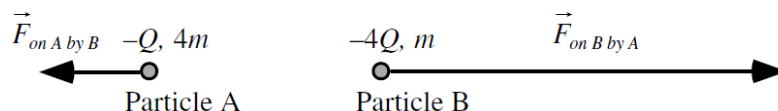
A student makes the following statement:

“Since $F = kQq/x^2$ and Q and q are both negative, the force on Q will be positive. Therefore, the force on Q points to the right.”

What, if anything, is wrong with this statement? If something is wrong, explain the error and how to correct it. If the statement is valid, explain why.

D1-WWT11: TWO NEGATIVELY CHARGED PARTICLES—FORCE

A student’s diagram for the electric forces acting on two negatively charged ($-Q$ and $-4Q$) particles is shown. Particle A has four times the mass of particle B.



What, if anything, is wrong with this diagram? If something is wrong, explain the error and how to correct it. If the diagram is valid, explain why.

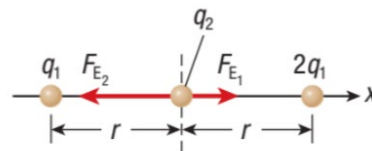
The Superposition Principle

Figure 3 The total force on q_2 equals the sum of the forces from charges q_1 and $2q_1$.

Example 1: Applying Coulomb's Law in One Dimension

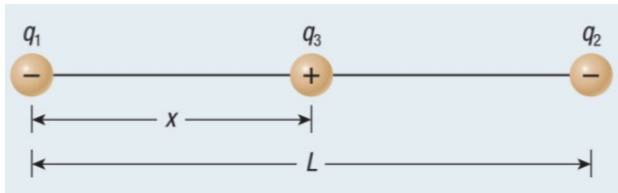
An early model of the hydrogen atom depicted the electron revolving around the proton, much like Earth revolving around the Sun. The proton and electron both have mass, so they exert a gravitational force upon each other. They also have a charge, so they exert an electric force on each other.

(a) The distance between the electron and the proton in a hydrogen atom is $5.3 \times 10^{-11} \text{ m}$, the charge of each is $1.6 \times 10^{-19} \text{ C}$, the mass of the electron is $9.11 \times 10^{-31} \text{ kg}$, and the mass of the proton is $1.67 \times 10^{-27} \text{ kg}$. Calculate the ratio of the electric force F_E to the gravitational force F_g .

(b) Determine the accelerations of the electron caused by both the electric force and the gravitational force.

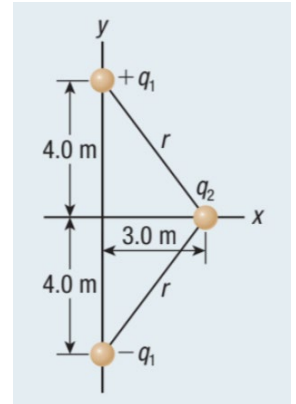
Example 2: Determining Electrostatic Equilibrium

Two charges, $q_1 = -2.00 \times 10^{-6} \text{ C}$ and $q_2 = -1.80 \times 10^{-5} \text{ C}$, are separated by a distance, L , of 4.00 m. A third charge, $q_3 = +1.50 \times 10^{-6} \text{ C}$, is placed somewhere between q_1 and q_2 , as shown in the figure below, where the net force exerted on q_3 by the other two charges is zero. Determine the location of q_3 .



Example 3: Applying Coulomb's Law in Two Dimensions

Two point particles have equal but opposite charges of $+q_1$ and $-q_1$. The particles are arranged as shown in the figure on the right. Suppose a charge q_2 is placed on the x -axis as shown. $q_1 = 5.0 \times 10^{-6} \text{ C}$, $q_2 = 1.0 \times 10^{-6} \text{ C}$, and the distance between $+q_1$ and $-q_1$ is 8.0 m measured vertically along the y -axis. Calculate the magnitude and the direction of the net electric force on q_2 .



Lesson Reflection

State one thing you learned in this lesson:

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State one thing you need to go over:

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Check Your Understanding – Lesson 7.2

1. What does the term *point charge* assume? What's its importance?
2. What is the main difference between gravity and the electric force is direction?
3. Consider the forces produced between two objects. Is this true: the one acting on the object with less charge has a greater effect than the force acting on the object with a greater charge.

7.3 Electric Field

Lesson Goal: Understand the properties of an electric field

Success Criteria:

- ✓ define and draw an electric field and a uniform electric field
- ✓ define a test charge and explain its role in electric fields
- ✓ differentiate between a point charge and a test charge
- ✓ demonstrate the use of a test charge in determining direction of electric force and electric field
- ✓ calculate the electric field strength for a point charge at a given distance
- ✓ describe an electric dipole
- ✓ give technological and natural examples of electric fields

Go through presentations [PP7 \(Electric Fields\)](#) let's look at some examples.

Electric Field ($\vec{E}$) the region in which a force is exerted on an electric charge; the electric force per unit positive charge. The unit is N/C

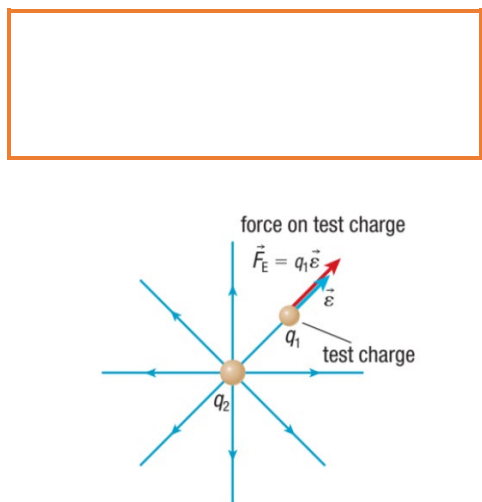
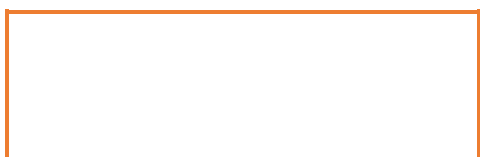


Figure 2 The electric field at a particular point in space is related to the electric force on a test charge q_1 at that location.



Electric Field Lines

Now that we went through [PP8 \(Electric Field Lines\)](#) let's go through some example.

An electric field exists in a region around a charge. The field and its properties can be represented with electric field lines.

Electric field lines the continuous lines of force around charges that show the direction of the electric force at all points in the electric field.

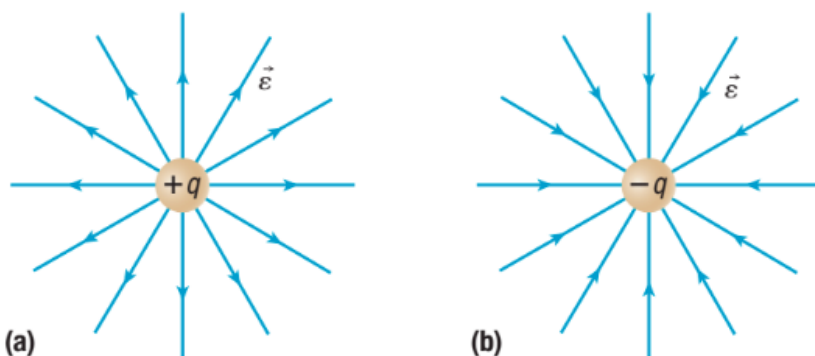
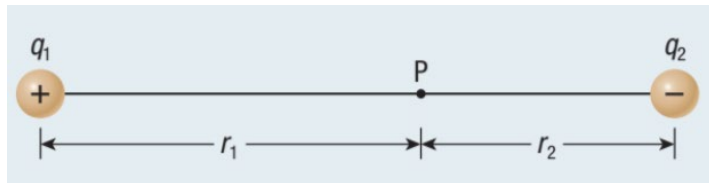


Figure 6 Electric field lines near a point charge placed at the origin. (a) If the charge is positive, the electric field lines are directed outward, away from the charge. (b) If the charge is negative, the electric field lines are directed inward, toward the charge.

Example 1: Electric Field Due to Two Point Charges in One Dimension

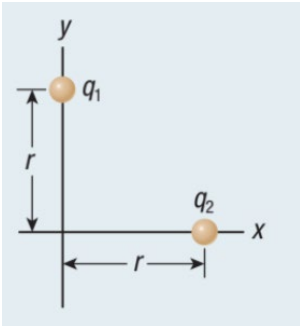
Two point charges are 45 cm apart (refer to the figure below). The charge on q_1 is 3.3×10^{-9} C, and the charge on q_2 is -1.00×10^{-8} C.



- Calculate the net electric field at point P, 27 cm from the positive charge, on the line connecting the charges.
- A new charge of $+2.0 \times 10^{-12}$ C is placed at P. Determine the electric force on this new charge.

Example 2: Electric Field Due to Two Point Charges in Two Dimensions

Two point charges are arranged as shown in the figure below. $Q_1 = 4.0 \times 10^{-6} \text{ C}$, $q_2 = -2.0 \times 10^{-6} \text{ C}$, and $r = 3.0 \text{ cm}$. Calculate the magnitude of the electric field at the origin.



How to Produce Uniform Electric Fields

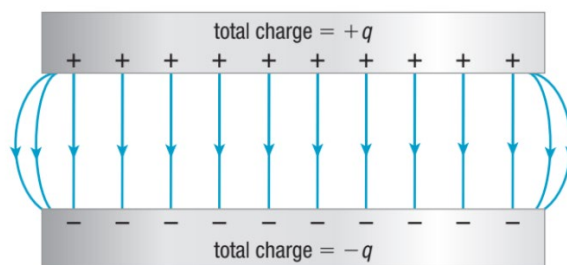


Figure 12 The electric field between two parallel conducting plates is uniform in direction and magnitude.

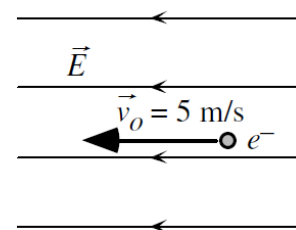
- The two “infinite” parallel plates.
- The separation between the plates is much smaller than their surface area
- One plane has a positive charge, and the other plane has a negative charge.
- The charge spreads uniformly along each plane.
- The electric field lines connecting two unlike charges extends straight from the positive to the negative charge.
- The electric field between the planes of charge extends from the positive plane of charge to the negative plane and is uniform.
- These field lines are straight, parallel to each other, and perpendicular to the planes of charge.
- At any location between the planes, the electric field has the same magnitude and direction.
- Outside the planes, the vector sum of the electric fields from all the individual charges in the two parallel planes yields a value of zero. In short, the electric field outside the parallel plates is zero.
- Except near the edges of the plates, the magnitude of the electric field depends only on the amount of charge, the area of the plates, and the material between the plates.

D1-WWT32: ELECTRON IN A UNIFORM ELECTRIC FIELD—VELOCITY

An electron is placed in a uniform electric field with an initial velocity of 5 m/s as shown. A student makes the following statement:

“The electron will continue to move in the same direction at a constant velocity because it is moving in the same direction as the electric force on it; since the electric field is constant, the force on the electron is constant.”

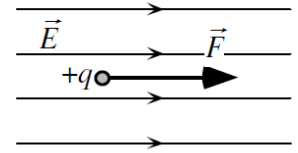
What, if anything, is wrong with this statement? If something is wrong, explain the error and how to correct it. If the statement is valid, explain why.



D1-LMCT33: POSITIVE CHARGE IN A UNIFORM ELECTRIC FIELD—ELECTRIC FORCE

A particle with a charge $+q$ is placed in a uniform electric field.

Identify from choices (i)–(vi) how each change described in (a) to (e) will affect the electric force on the particle.



This change will:

- (i) change only the **direction** of the electric force.
- (ii) **increase** the magnitude of the electric force.
- (iii) **decrease** the magnitude of the electric force.
- (iv) **increase** the magnitude and change the **direction** of the electric force.
- (v) **decrease** the magnitude and change the **direction** of the electric force.
- (vi) **not affect** the electric force.

All of these modifications are changes to the initial situation shown in the diagram.

- a) The charge q on the particle is doubled. ____ Explain your reasoning.
- b) The sign of the charge q on the particle is changed to the opposite sign. ____ Explain your reasoning.
- c) The particle is given a push, causing a leftward initial velocity. ____ Explain your reasoning.
- d) The magnitude of the uniform electric field is halved. ____ Explain your reasoning.
- e) The direction of the uniform electric field is rotated 90° clockwise. ____ Explain your reasoning.

Lesson Reflection

State one thing you learned in this lesson:

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State one thing you need to go over:

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7.4 Potential Difference and Electric Potential

Lesson Goal: Express the electric field in terms of energy per charge

Success Criteria:

- ✓ determine the change in electric potential energy of a
- ✓ charge if given the electric field, distance moved, and quantity of charge
- ✓ describe the changes in energy for work done against the electric field and work done with the electric field
- ✓ define and use electric potential in problems
- ✓ explain how the magnitude of the electric field varies with
- ✓ the electric potential difference and the change in position in the field

Now that we went through the presentation [PP9 \(Electric Potential Energy\)](#) let's look at some examples.

Work and Electric Potential Difference

Electric Potential Energy (E_E) the energy stored in a system of two charges a distance Δd apart, or the energy stored in an electric field that can do work on a positively charged particle.

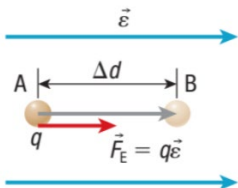


Figure 2 The electric field gives rise to an electric force, which moves in the same direction as the charge if the charge is positive.



Example 1: Potential Energy Difference in an Electric Field

A charged particle moves from rest in a uniform electric field.

(a) For a proton, calculate the change in electric potential energy when the magnitude of the electric field is 250 N/C, the starting position is 2.4 m from the origin, and the final position is 3.9 m from the origin.

(b) Calculate the change in electric potential energy for an electron in the same field and with the same displacement.

(c) Calculate the change in electric potential energy for an electron accelerated in an electric field with the same magnitude but opposite direction as in (a) and (b), and with a starting position of 2.4 m from the origin and a final position of 5.0 m from the origin.

Example 2: Dynamics of Charged Particles

- (a) Using the law of conservation of energy, calculate the speed of the proton in part (a) of **Example 1** for the given displacement. Assume that the proton starts from rest.
- (b) Determine the initial speed of the electron in part (b) of **Example 1**, assuming its speed has decreased to half of its initial speed after the same displacement, Δd .

Electric Potential

Electric Potential the value, in volts, of potential energy per unit positive charge for a given point in an electric field. The unit, volts is $1 \text{ V} = 1 \text{ J/C}$. In other words, electric potential is a measure of how much electric potential energy is associated with a specific quantity of charge at a particular location in an electric field.



Electric Potential Difference (ΔV) the amount of work required per unit charge to move a positive charge from one point to another in the presence of an electric field



Good explanation - <https://howtomechatronics.com/learn/electric-potential-electric-potential-difference-voltage/>

Example 3: TV Tubes and Particle Accelerators

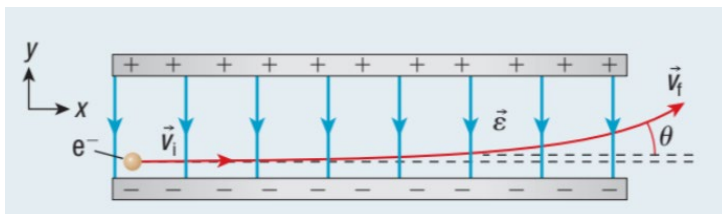
The cathode-ray tubes in old television sets and computer monitors work in a way that is similar to certain parts in particle accelerators. Both devices accelerate particles in a similar way, using the uniform electric field between conducting plates. Looked at another way, the particles accelerate as they move through an electric potential difference.

(a) An electron leaves the negative plate of a cathode-ray tube and travels toward the positive plate. The electric potential difference between the plates is 1.5×10^4 V. Using the law of conservation of energy and the definition of electric potential difference, calculate the speed of an electron as it reaches the positive plate in a cathode-ray tube. Assume that the electron is initially at rest. The mass of an electron is 9.11×10^{-31} kg.

(b) Calculate the magnitude of the electric field at a distance of 15 cm, which is at the end of the cathode-ray tube.

Example 4: Calculating the Speed of a Deflected Charged Particle

An electron moves horizontally with a speed of 1.6×10^6 m/s between two horizontal parallel plates. The plates have a length of 12.5 cm, and a plate separation that allows a charged particle to escape even after being deflected (refer to the figure below). The magnitude of the electric field within the plates is 150 N/C. Calculate the final velocity of an electron as it leaves the plates.



Lesson Reflection

State one thing you learned in this lesson:

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State one thing you need to go over:

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7.5 Electric Potential and Electric Potential Energy Due to Point Charges

Lesson Goal: Calculate electric potential due to point charge and the electric potential energy due to two charges

Success Criteria:

- ✓ calculate electric potential at a distance r from a point charge
- ✓ describe the total electric potential at a point P for a system of charges, which equals the algebraic sum of potentials at P due to each individual charge at a distance that separates the charge and P
- ✓ show the work done to move a test charge through an electric field, which equals the change in electric potential energy

Now that we went through the presentation [PP10 \(Electric Potential Difference\)](#) let's work through some examples.

Electric potential due to a point charge the electric potential is inversely proportional to the distance from the charge and proportional to the amount of charge producing the field

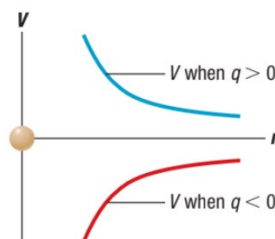
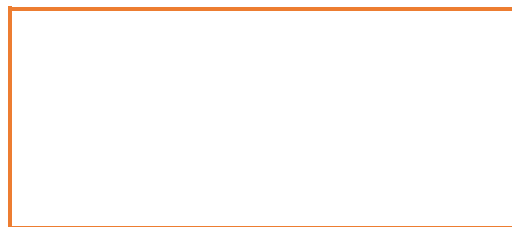


Figure 2 Electric potential near a point charge. The blue curve is for a positive point charge, and the red curve is for a negative point charge.

Note: The electric potential energy depends on the position d of the test charge in an electric field $\vec{E}$. If the charge producing the electric field has the same sign as the test charge q , the work done by the electric field on q is negative. This means that $\Delta E_E > 0$ for $\Delta d > 0$, and the sign of potential energy (and thus potential) is reversed from that of the uniform electric field.

Note: Only changes in potential energy are important, so you must always be clear about choosing the reference point for potential energy. Electric potential is proportional to electric potential energy, so you must also pay attention to the reference point for electric potential. For example, when dealing with a point charge, follow the standard convention by choosing $V=0$ at an infinite distance from the source charge. Infinite distance represents a distance so far away that any potential produced by a charge is negligible and for all intents and purposes equal to zero. In many other problems, choose Earth as $V=0$ because Earth conducts charge well. Put another way, the electric ground is the point where $V=0$.

Electric Potential Energy of Two Point Charges

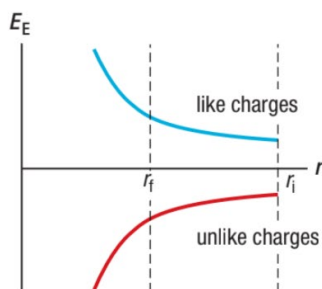
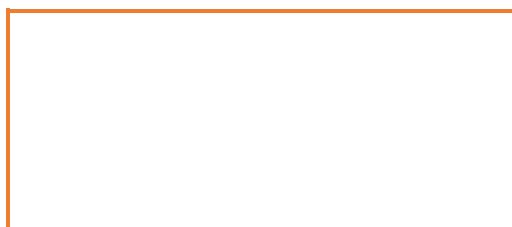


Figure 3 Electric potential energy as a function of the separation r between two charges q_1 and q_2 . The blue curve is for a repulsive force, and the red curve is for an attractive force.

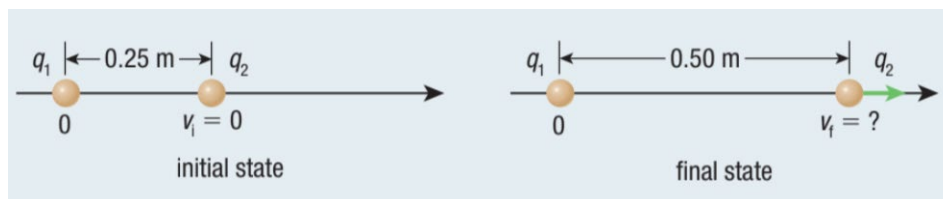
Example 1: Calculating the Electric Potential

A point charge with a charge of $4.0 \times 10^{-8} \text{ C}$ is 4.0 m due west from a second point charge with a charge of $-1.0 \times 10^{-7} \text{ C}$.

- a) Calculate the total electric potential due to these charges at a point P, 4.00 m due north of the first charge.
- b) Calculate the work required to bring a third point charge with a charge of $2.0 \times 10^{-9} \text{ C}$ from infinity to point P.

Example 2: Electric Potential Energy and Dynamics

A point charge q_1 with charge $2.0 \times 10^{-6} \text{ C}$ is initially at rest at a distance of 0.25 m from a second charge q_2 with charge $8.0 \times 10^{-6} \text{ C}$ and mass $4.0 \times 10^{-9} \text{ kg}$ (see figure below). Both charges are positive. Charge q_1 remains fixed at the origin, whereas q_2 travels to the right upon release. Determine the speed of charge q_2 when it reaches a distance of 0.50 m from q_1 .



Example 3: A Head-on “Collision”

Two particles, a proton with charge 1.60×10^{-19} C and mass 1.67×10^{-27} kg and an alpha particle (helium-4 nucleus) with charge 3.20×10^{-19} C and mass 6.64×10^{-27} kg, are separated by an extremely large distance. They approach each other along a straight line with initial speeds of 3.00×10^6 m/s each. Calculate the separation between the particles when they are closest to each other.

Lesson Reflection

State one thing you learned in this lesson:

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State one thing you need to go over:

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7.6 The Millikan Oil Drop Experiment

Lesson Goal: Understand the significance of the Millikan Oil drop experiment and

Success Criteria:

- ✓ describe how Robert Millikan determined the magnitude of the electron's charge
- ✓ state the value of the charge on an electron and that it is the same value as the elementary charge
- ✓ explain the significance of the elementary charge within the context of subatomic particles

Now that we went through the presentation [PP11 \(The Millikan Oil Drop Experiment\)](#)

Elementary charge is the magnitude of the electric charge carried by a proton, equal to the absolute value of the electric charge of an electron.

Videos:

1. <https://www.youtube.com/watch?v=UFiPWv03f6g>
2. <https://www.youtube.com/watch?v=ijHKu6iXIRk&t=3s>
3. [Millikan's oil drop experiment to determine charge of an electron](#)
4. [Millikan's Oil Drop Experiment - A Level Physics](#)
5. [Millikan's oil drop experiment explained](#)

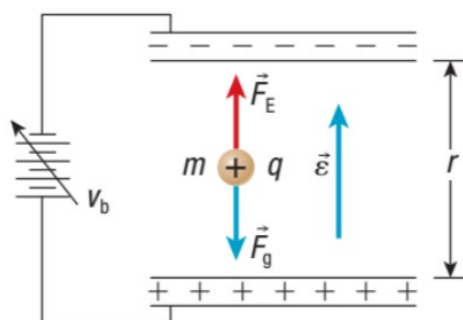


Figure 2 A charged oil drop will be suspended between two charged plates when the electric force on the drop balances the gravitational force.

Example 1: Calculating the Charge on an Object

Calculate the charge on a small sphere with an excess of 3.2×10^{14} electrons.

Example 2: The Elementary Charge and an Oil Drop

In a Millikan-type experiment, two horizontal plates maintained at a potential difference of 360 V are separated by 2.5 cm. A latex sphere with a mass of 1.41×10^{-15} kg hangs between the plates, the upper plate of which is positive.

- a) Is the sphere negatively or positively charged?
- b) Calculate the magnitude of the charge on the latex sphere.
- c) Determine the number of excess or deficit particles on the sphere.

Lesson Reflection

State one thing you learned in this lesson:

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State one thing you need to go over:

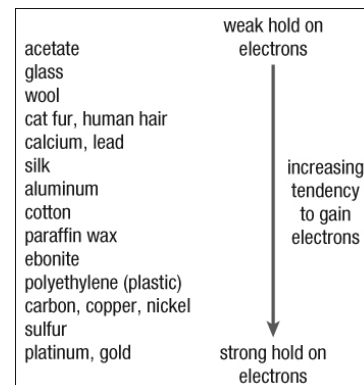
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Chapter 7 - Check Your Knowledge – Definitions

Term	Definition
___ elementary charge (e)	1. a measurable natural value that never varies and can be determined by experimentation
___ electric potential	2. the amount of work required per unit charge to move a positive charge from one point to another in the presence of an electric field
___ electric dipole	3. the energy stored in a system of two charges a distance Δd apart, or the energy stored in an electric field that can do work on a positively charged particle
___ electric field	4. the proportionality constant in Coulomb's law; equals to $8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$
___ electric potential due to a point charge	5. any substance in which electrons are able to move easily from one atom to another
___ superposition principle	6. the continuous lines of force around charges that show the direction of the electric force at all points in the electric field
___ insulator	7. the value, in volts, of potential energy per unit positive charge for a given point in an electric field
___ electric potential difference	8. the magnitude of the electric charge carried by a proton, equal to the absolute value of the electric charge of an electron
___ fundamental physics constant	9. the SI unit of electric charge; the symbol is C
___ electric field lines	10. the region in which a force is exerted on an electric charge; the electric force per unit positive charge; it is measured in N/C
___ Coulomb's constant	11. any substance in which electrons are not free to move easily from one atom to another
___ Coulomb	12. a pair of equal and opposite electric charges with centers separated by a small distance
___ conductor	13. the resultant, or net, vector acting at a given point equals the sum of the individual vectors from all sources, each calculated at a given point
___ electric potential energy	14. the electric potential is inversely proportional to the distance from the charge and proportional to the amount of charge producing the field

Chapter 7 Communication Questions

1. Refer to the electrostatic series. A carbon rod is rubbed with silk. Describe the charge transfer that occurs.
2. How does grounding electrical appliances protect users?
3. Refer to the series of three diagrams. Explain the type of charging taking place, and why it works.
4. Refer to the series of three diagrams. Explain the type of charging taking place, and why it works.
5. Anton says, “When I bring a positively charged rod close to a metal sphere, the negative charges in the sphere move toward the rod, and the positive charges in the sphere move away from the rod.” Is Anton correct?
6. Two identical electric charges q are placed on two balloons whose centres are separated by a distance r . If the charges and distance are all doubled in value, what change is apparent in the electric force between the balloons?
7. Two identical positive charges Q are separated a distance r apart along a northeast-southwest line. A positive test charge q is placed such that it is north of one charge and west of the other. In what direction does the net electric force on the third charge point? Explain your answer.
8. What are the requirements for the test charge q that is used to measure an electric field? Why are these requirements necessary?
9. How could you construct an apparatus that would create a uniform electric field? Indicate any restrictions on using your apparatus.
10. What evidence is there that an electric potential can exist in empty space?



11. Use a graphic organizer to compare the properties of the electron and the proton.
12. Although quarks seem to have fractional elementary charges, the charge on the proton is still considered as the elementary charge. Explain this discrepancy.
13. In a science fiction story by George O. Smith, inhabitants of a space station protect themselves against invaders by building an electron gun, and shooting a beam of electrons against their antagonists. After a while, the electron gun no longer seems to work very well. Think about what you have learned in this chapter, and suggest a reason for this.
14. Think about a device that you use every day that depends on electric fields for its operation. Indicate what the device is, and why it is important to you.
15. Describe how electrostatic precipitators work.
16. Should electrostatic precipitators be mandatory for use? Consider a source of air pollution near where you live. Provide reasons for and against to the polluter to make use of electrostatic precipitators.
17. What is an electric field mill? Describe how it works.
18. Sometimes a change in technology can occur quickly. Around 2000, most televisions and computer monitors used electric fields in cathode ray tubes. Now, they use electric fields in LCD screens. Reflect on what you have learned about cathode ray tubes and LCD screens. What kinds of problems arise when there is a rapid change in technology such as this?
19. Reflect on what you have learned about electric fields in this chapter. What evidence from your own experience do you have that parts of your body are affected by electric fields?
20. How would you explain the difference in electrical potential between that of a car battery and that of an outlet in your home to someone who understands work and energy, but does not understand electricity.

